Study Report: Generative AI with LangChain - Generative AI Engineer

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Family:	Generative AI Engineer
Level:	Intermediate
Chapters:	7
Source:	Reference: Generative AI with LangChain - Ben Auffarth & Laura Trotta

Official sources and free libraries

- https://www.packtpub.com/en-us/product/generative-ai-with-langchain-9781835084848

Summary

Study report for *Generative AI with LangChain* by Ben Auffarth & Laura Trotta (Intermediate level) mapped to the Generative AI Engineer role. LangChain chains, RAG, agents, and LLM application structure.

Chapters

Track: Book-Intro

Chapter 1: About Generative AI with LangChain [Intermediate]

Chapter focus: About Generative AI with LangChain *(Level: Intermediate)**

This study report summarizes *Generative AI with LangChain* by Ben Auffarth & Laura Trotta for the Generative AI Engineer role. The resource is rated Intermediate level. LangChain chains, RAG, agents, and LLM application structure. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Generative AI with LangChain
# Author: Ben Auffarth & Laura Trotta
# Role: Generative AI Engineer
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on Generative AI with LangChain.

When to use it
When learning Generative AI Engineer skills at Intermediate level.

Where to use it
Generative AI Engineer training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role *(Level: Intermediate)**

Generative AI Engineer professionals use ideas from Generative AI with LangChain to solve real workplace problems. LangChain chains, RAG, agents, and LLM application structure. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Generative AI Engineer
Book focus: LangChain chains, RAG, agents, and LLM application structure.
Recommended level: Intermediate
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Generative AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered** *(Level: Intermediate)

The main topics in Generative AI with LangChain include practical concepts described as: LangChain chains, RAG, agents, and LLM application structure. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Generative AI Engineer jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: RAG Pipeline Architecture [Intermediate]

Chapter focus: RAG Pipeline Architecture** *(Level: Intermediate)

RAG = Retrieve relevant chunks ? Augment prompt ? Generate answer. Components: ingest, chunk, embed, index, retrieve, rerank, generate, cite. Failure modes: wrong retrieval, hallucination despite context, stale index.

Code Reference:

```
# RAG flow
# query -> embed -> top_k retrieve -> build prompt with context -> LLM -> answer + cites
```

What it shows: Explicit flow diagram helps debug which stage failed.

Topic guide

What it actually is
RAG grounds LLM answers in private or fresh documents.

When to use it
Enterprise knowledge assistants and support bots.

Where to use it
Legal, HR policy, and technical documentation copilots.

Real use example
Teacher asks about grading rubric - RAG pulls 2025 PDF not outdated blog.

Chapter 5: Applied: Chunking and Metadata Strategies [Intermediate]

Chapter focus: Chunking and Metadata Strategies *(Level: Intermediate)**

Chunk size 500-1000 tokens with 10-20% overlap preserves continuity. Attach metadata (source, page, section) for citations and filtered retrieval.

Code Reference:

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
splitter = RecursiveCharacterTextSplitter(chunk_size=800, chunk_overlap=120)
chunks = splitter.split_documents(docs)
```

What it shows: Recursive splitter respects paragraph boundaries better than fixed char cuts.

Topic guide

What it actually is
Smart chunking balances retrieval precision and context completeness.

When to use it
Indexing long PDFs, wikis, and API docs.

Where to use it
Any RAG system over heterogeneous documents.

Real use example
Smaller chunks improved recall on API reference; larger chunks helped narrative guides.

Chapter 6: Applied: Vector Databases: Chroma, Pinecone, pgvector [Intermediate]

Chapter focus: Vector Databases: Chroma, Pinecone, pgvector *(Level: Intermediate)**

Chroma is local/dev-friendly; Pinecone managed scale; pgvector keeps vectors in Postgres. Use namespaces per tenant; HNSW indexes trade memory for speed.

Code Reference:

```
# pgvector
CREATE EXTENSION vector;
```

```
CREATE TABLE chunks (id serial, content text, embedding vector(1536));
CREATE INDEX ON chunks USING hnsw (embedding vector_cosine_ops);
```

What it shows: HNSW index accelerates similarity search on millions of vectors.

Topic guide

What it actually is
Vector DBs persist embeddings with fast approximate nearest neighbor search.

When to use it
Production RAG at scale with filtered metadata queries.

Where to use it
SaaS copilots, semantic search, and recommendation recall.

Real use example
pgvector co-locates vectors with OLTP data - simpler ops for mid-size teams.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice *(Level: Intermediate)**

Finish Generative AI with LangChain with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Generative AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it
CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.